

Lecture 3

image sampling and image acquisition

Fons J. Verbeek. Image Analysis Spring 2006

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1

Content

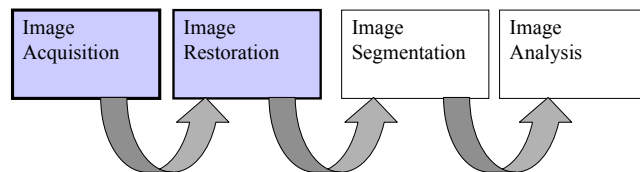
- Optical system
- Sampling, Sampling density
- Reconstruction of a signal
- Restoration of a signal
- Sensor system
- Measurements on the sensor system

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2

Sequence of Processing

- There is a standard sequence of processing and analyzing images.



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3

Images

- Image Processing:
 - Image in → Image out
- Picture Processing:
 - Explicitly to enhance the photographic output
- Image Analysis:
 - Image in → Numbers out
- Visualization, Computer Graphics
 - Numbers in → Image out

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4

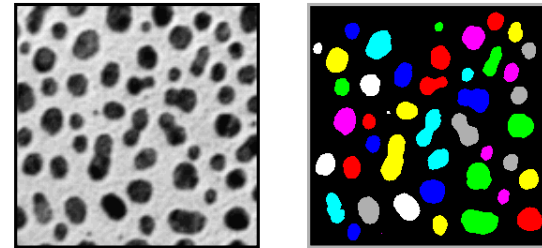
Image Elements

- Pixel
 - Smallest addressable element in an image;
 - Addressed by row and column in a matrix
 - An element in a CCD-array or CCD-well.
- Voxel
 - Element in a volumetric image;
 - Addressed by row, column and plane in a 3D matrix.
- Imel
 - Texel texture element, typical in texture analysis
 - Tixel image element changing over time (4D)
 - Scenel a scene element
 - Dixel a display element

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5

Image Analysis



- To do the measurements how should a scene be sampled?
- What acquisition system is required to do that?

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... So what about digital images

- Optical system (research microscope)
- ...lens produces
- ...image; is project on
- ...sensor (CCD) and
- ...convert (ADC) to digital representation

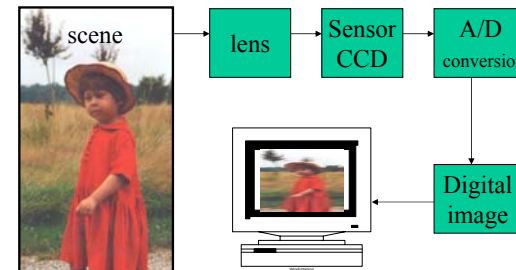
typical flow for quantitative light microscopy

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7

Digital Imaging System

- Lens system
- Projection system
- Illumination system
- Sensor system (camera)
- Display system (monitor)

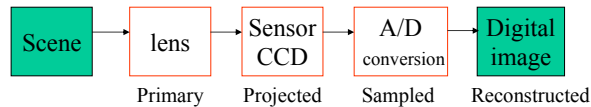


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8

Digital Imaging Microscope

- Lens system
- Projection system
- Illumination system
- Sensor system (camera)
- Display system (monitor)



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9

Sampling

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10

Resolution and Digitization

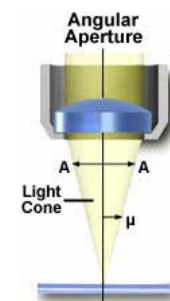
- The resolution of an optical system is determined by the lens
 - Abbe diffraction limit
 - Definition
- The resolution of an acquisition system for electronic imaging is determined by the combination
 - Lens, numerical aperture
 - Sensor, CCD characteristics
 - digitizer (ADC), frame-grabber characteristics

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11

Sampling theory

- Optical systems are band-limited.
 - The lens aperture determines what frequencies are passed in the system.
 - Optical systems are well described with Fourier Theory
- Sampling theorem
 - Abbe (diffraction limit)
 - Rayleigh
 - Nyquist
 - Shannon



$$N.A. = n \sin \mu$$

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12

Contributors to Diffraction Theory



Joseph von Fraunhofer
(1797-1826)



Sir George Airy
(1801-1892)



Ernst Abbe
(1840-1905)



Lord Rayleigh (John Strutt)
(1842-1919)

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13

Abbe diffraction

- A detail with a particular spacing in the specimen is resolved when the NA of the objective lens is large enough to capture the first order diffraction pattern produced by the wavelength employed.
- This is related to the Airy Disk (telescope)
- This is the cutoff frequency of the lens. See also Nyquist criterion.

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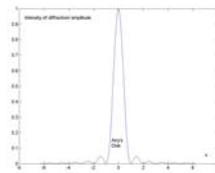
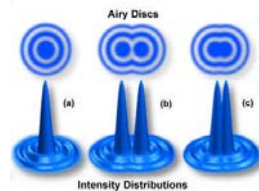
14

The Rayleigh Criterion

- For a circular aperture the maximum spatial resolution of a system is given by:

$$R = \frac{0.61 * \lambda}{NA}$$

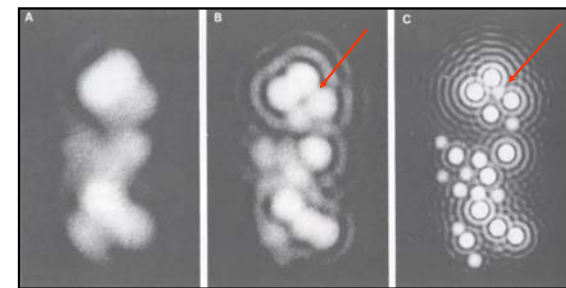
- Resolved = Separated
- Point light sources can be resolved, if a minimal distance between the peaks equals the radius of the central disk.



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The Airy-Disk



Increasing NA at same magnification; resolving power

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16

Diffraction Limits Circular Aperture

- With a given $\lambda = 500\text{nm}$, 550nm this translates into:

Objective	NA	Rayleigh		Rayleigh	
		500 nm	microns	550 nm	microns
5	0,15	2,03333E-06	2,03	2,23667E-06	2,24
10	0,25	0,00000122	1,22	0,000001342	1,34
20	0,5	0,00000061	0,61	0,000000671	0,67
25	0,6	5,08333E-07	0,51	5,59167E-07	0,56
40	0,85	3,58824E-07	0,36	3,94706E-07	0,39
63	1,35	2,25926E-07	0,23	2,48519E-07	0,25

- NA and wave-length (λ) contribute to resolving power
- smaller wave-length results in higher resolving power

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17

Incoherent vs. Coherent Imaging

- Incoherent imaging:
 - phase relationship from light emitted by different specimen parts varies randomly
 - typical for fluorescent imaging
 - thus $R = \frac{0.61 * \lambda}{NA}$ does not hold in all cases
- Coherent imaging:
 - each point of the light source illuminated the specimen
 - typical for trans-illumination and reflective microscopy
 - this transforms Rayleigh into: $\frac{0.6 * \lambda}{NA} \leq R \leq \frac{0.8 * \lambda}{NA}$

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Resolution for Coherent light

- With a given $\lambda = 550\text{nm}$ this translates into this range:

Objective	NA	Rayleigh	lower	Rayleigh	upper
		550 nm	microns	550 nm	microns
5	0,15	0,0000022	2,2	2,93333E-06	2,93
10	0,25	1,32E-06	1,32	0,00000176	1,76
20	0,5	6,6E-07	0,66	0,00000088	0,88
25	0,6	5,5E-07	0,55	7,33333E-07	0,73
40	0,85	3,882E-07	0,39	5,17647E-07	0,52
63	1,35	2,444E-07	0,24	3,25926E-07	0,33

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19

Characterizing Imaging properties

- Resolution by Rayleigh, lower bound.
- Point Spread Function
 - More complete description (using Bessel Functions)
$$psf(r) = \left(\frac{2J_1(v)}{v} \right)^2 \quad v = \frac{2\pi}{\lambda} r \sin \beta$$
 - psf is measured using test objects that have a point like behavior
 - Modulation Transfer Function: MTF = Fourier (psf)
 - MTF characterizes imaging system by relating the objective to spatial frequency

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20

Nyquist & Shannon on Sampling

Theorem of sampling

- Formulated by Harry Nyquist; "Certain topics in telegraph transmission theory," Trans. AIEE, vol. 47, Apr. 1928.
- Mathematically proven by Claude Shannon; Founder of modern information theory
- Also attributed to Vladimir Kotelnikov (proof)



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Shannon Sampling Theorem

- Shannon Sampling theorem
 - To preserve the spatial resolution of the original image, the digitizing device must use a sampling interval no greater than one-half the size of the smallest resolvable feature of the optical image.
- This relates the the Nyquist Criterion

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Nyquist sampling theory

- Nyquist theory describes the sampling frequency (f) required to represent the true identity of the sample.
- i.e., how many times an image must be sampled to know that a sample truly represents the image?
- Nyquist stated that to capture the periodic components of frequency f in a signal we need to sample at least $2f$ times.

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Nyquist Criterion

- In images: spatial frequency is the cycle rate between brightness transitions.
- So, in essence, you must sample at least 2 times the highest spatial frequency.
- The result is that spatial frequencies in a band limited signal are properly represented in your digitized image.

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24

Reoccurring Concepts & Definitions

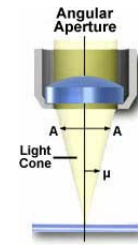
- Nyquist Frequency: highest frequency that occurs in the signal, aka Nyquist limit
- Sampling Frequency: at least twice the Nyquist frequency
- Nyquist Sampling Rate is identical to Sampling Frequency

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25

Sampling Theory & Microscope

- The lens acts like a low-pass filter with a cutoff frequency given by $w_{\max} = \frac{2 \times NA}{\lambda}$
- Where $NA = n \sin \mu$ and n = refractive index



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26

Aperture angle and RI

- Aperture angle from given NA and RI (air, water, oil)

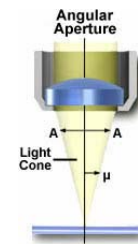
mag	NA	refractive index	radians	degrees
25	0,6	1	0,64	36,67
40	0,85	1	1,02	58,44
40	0,85	1,33	0,69	39,53
40	1,3	1,33	1,36	77,92
40	1,3	1,52	1,03	59,01

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27

Nyquist Criterion & Microscope

- Image has to be sampled above the Nyquist frequency:
 - i.e. twice the cut-off frequency.
- Thus: $SD_{Nyquist} = \left(\frac{4 * NA}{\lambda} \right)$
 - in pixel/μm
- Consequently, for
 - 25x (0.6) at λ = 500 nm → SD = 4.8 pixels/μm
 - 40x (1.3) at λ = 500 nm → SD = 10.4 pixels/μm (oil)
- And thus for
 - 40x * 1.3 we need at least 11 pixels/μm



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Sampling Density

- Sampling Density (SD) represents the number of pixels per unit length.
- Sampling Density is also referred to as the sampling interval.
- Smaller pixels (sensor elements) give rise to higher SD.
- Requires calibration with test pattern.

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29

Sampling Density: Rules

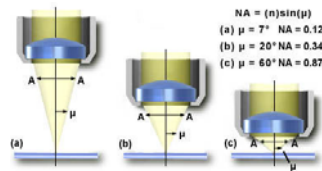
- Sometime, depending on image processing steps a higher SD is chosen.
- In general, sample above Nyquist criterion for image analysis
- Based on measurement CV (precision), Rules of thumb:
 - 50 samples per average cell diameter for area.
 - 25 samples per analogue length for length
 - 150 samples per average cell diameter for texture

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30

Factors of Resolution

- Numerical Aperture, aperture angle
- Wavelength
- Refractive index of the medium, oil vs. air
- Different approach for Processing vs. Analysis



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31

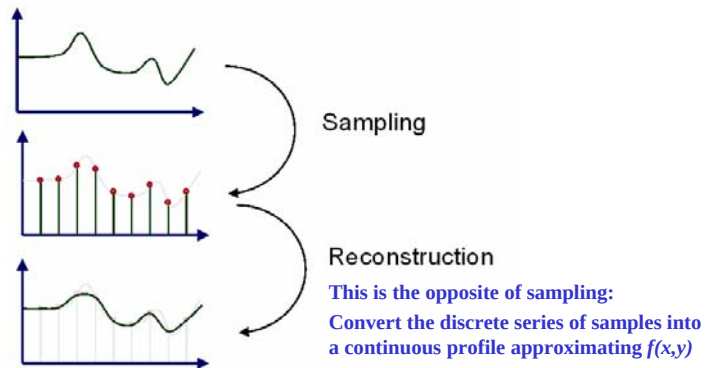
Image Reconstruction

- Image reconstruction is the process of converting data into continuous analog data.
- We want the signal to appear continuous.
- A band-limited signal can be reconstructed.
- Principles of proper sampling (Nyquist).
- Filters
 - Zero to higher-order
 - Band-limit filter to Nyquist frequency
 - Display medium
 - Human visual system

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32

Sampling & Reconstruction



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33

Aliasing

- Aliasing occurs when contribution from higher frequencies show up at lower frequencies.
- Pixels are too far apart compared to the high frequency pattern (detail) the image.
- In general: result of too few samples
- Aliasing may result in a Moiré-pattern
- Anti-aliasing applications/filters are used in the reconstruction phase
- Anti-aliasing especially required with abundance of high frequencies.

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34

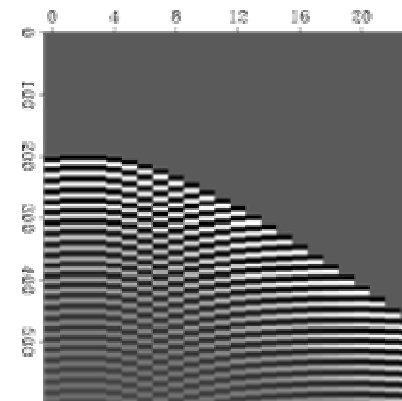
Aliasing Example: moiré pattern



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35

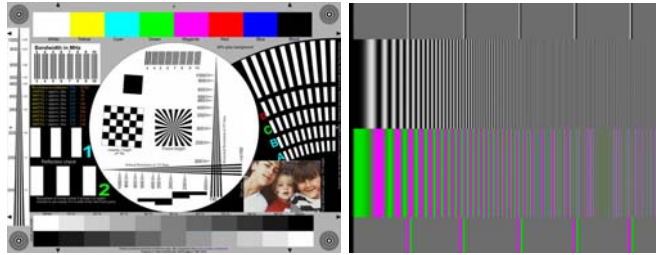
Aliasing Example: Artificial



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36

Aliasing Example: Video -TV

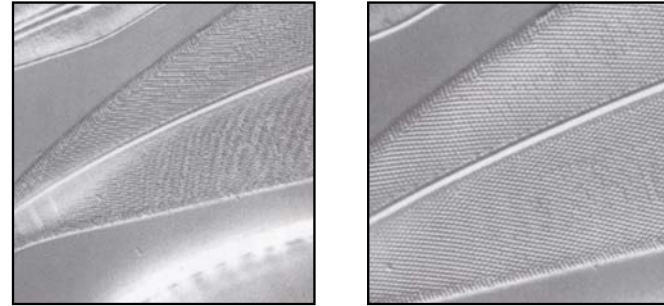


Test Patterns to tune anti-aliasing

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37

Aliasing Example: Microscopy



40x (0.9) lens

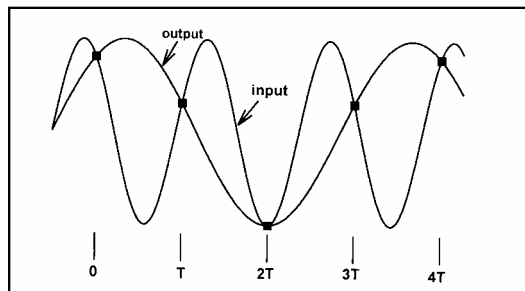
40x (0.9) + 1.5 relay lens

Diatom with very fine pattern

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38

Under-Sampling

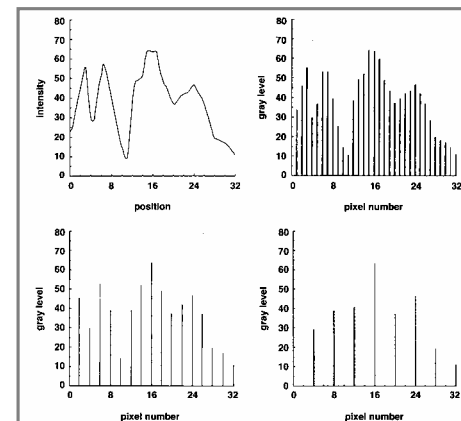


The number of samples is too little to "see" the detail ...

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39

Sampling Frequency



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40

Image Restoration

- The process of recovering an image from degradations that have occurred during creation and transmission of the image.
- Examples:
 - Noise stochastic
 - Blur deterministic
 - Motion deterministic
 - Sensor non-linearity deterministic
 - Lens characteristics deterministic
- If the source is known then the image can often be restored.

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41

Sensors (CCD)

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42

CCD

- Charged Coupled Device (CCD)
- Complementary Metal Oxide Semiconductor (CMOS)
- Integrated Circuits



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43

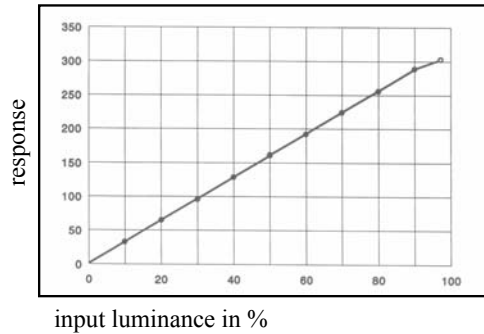
Sensor Characteristics

- Absolute Sensitivity: minimal amount of charge (photons) that can be detected.
- Relative Sensitivity: minimal amount of charge (photons) required to change output with 1 AD unit
- Spectral response: sensitivity of sensor to wavelengths
- Linearity: dose response, typical for CCD

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44

Linear response of CCD



In theory this is formulated with linear systems

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45

Monochrome video cameras

standard video camera

- CCD (charge coupled device)

scientific camera

- Progressive scan CCD (standard video)
- Slow Scan Camera (CCD)
- CID (charge injection device)
- HyperHAD CCD

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46

Image sizes & CCD sensor

standard video camera

- Depending on combination framegrabber
- Standard video format 768x574
- Standard image format 512x512

scientific camera

- progressive scan CCD (standard video)
- slow Scan Camera
 - 1024x1024
 - 2048x2048
 - 1300x1100

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47

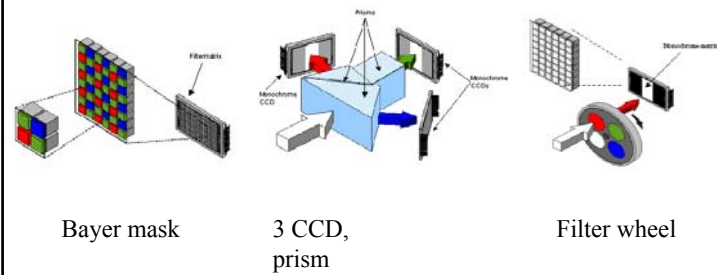
Color video cameras

- Interpolation (RGB sensitivity)
 - standard CCD, composite video
- 3 CCD sensors & prism
 - standard CCD 's, sensor alignment!
- Filter wheel & shutter
 - standard CCD - large area CCD (slow scan)
 - Luminance, Red, Green & Blue (band-pass)
- LCD shutter
 - large area CCD

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48

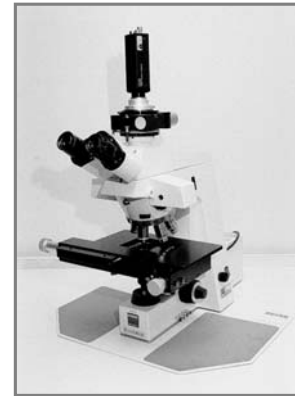
Color CCD camera Strategies



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49

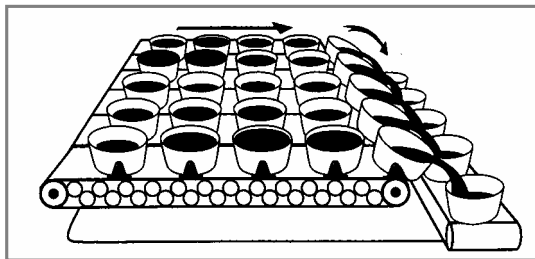
Image Acquisition Set Up



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50

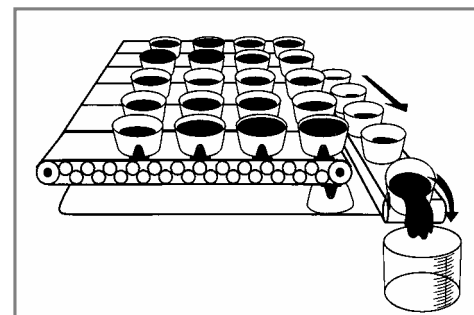
Parallel shift operation in CCD



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51

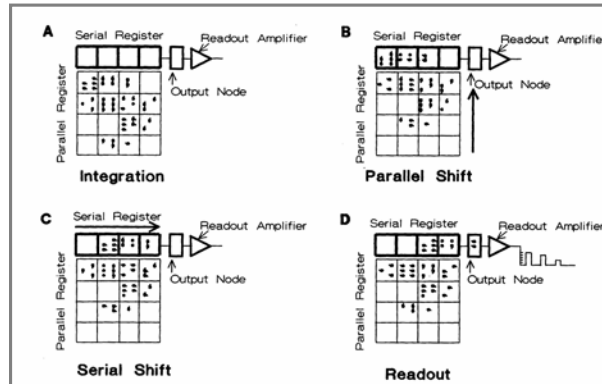
Serial shift and read-out in CCD



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52

Sequence of operations in CCD



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53

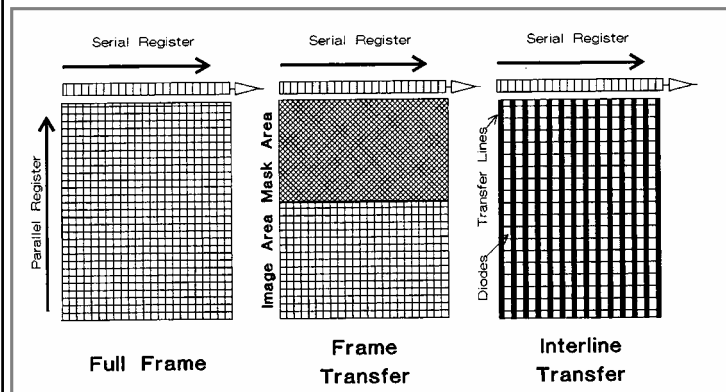
Standard CCD camera

- Allows integration on chip
- Averaging for noise suppression
- Newer CCD video camera's have square pixels: CCD wells
- Frame grabber does video sampling. Consequently, frame grabber hardware determines if pixels are sampled as squares.

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CCD sensor architecture



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Signal distortion on CCD (1)

- Photon noise: dual character of photons, Poisson distributed
- Dark current: Thermal agitation. Measured by taking a dark current image. Used to correct for Dark current (subtraction).
- Readout noise: Noise added at readout of wells. Low in scientific camera's.
- Quantization noise: Errors (roundoff) in conversion of charge to integer numbers.

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56

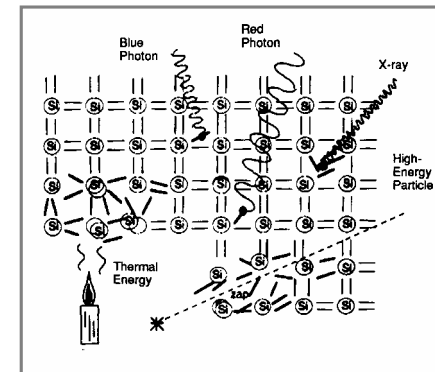
Signal distortion on CCD (2)

- Blooming: CCD well is saturated. Extra charge (electrons) passed to neighboring wells. Adjust exposure time.
- Smearing: If light reaches the CCD during read out
- Dead pixels: state of accuracy of the CCD.
 - The % of dead pixels is given.
 - The exact location of bad performing wells is given.

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57

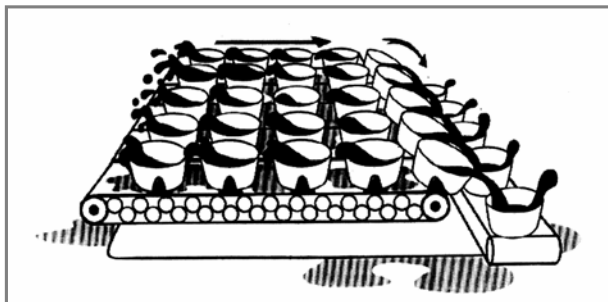
Noise on CCD



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58

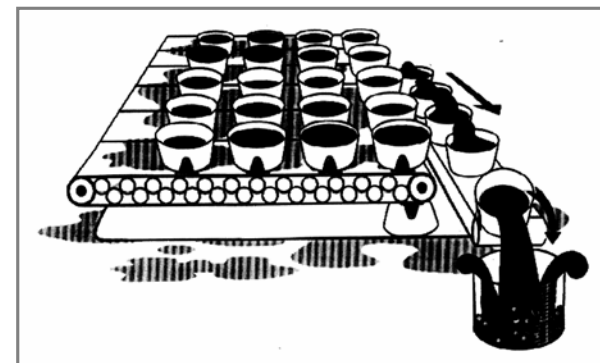
High speed parallel shift in CCD



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59

Charge loss in CCD read-out



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60

Noise reduction

standard video camera

- Averaging (N frames).
- Integration at low light level images, depends on frame grabber bus speed.

scientific camera

- Cooling of sensor (long integration)
- Binning: combine the charge in the CCD wells at read out. # pixels is reduced.

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61

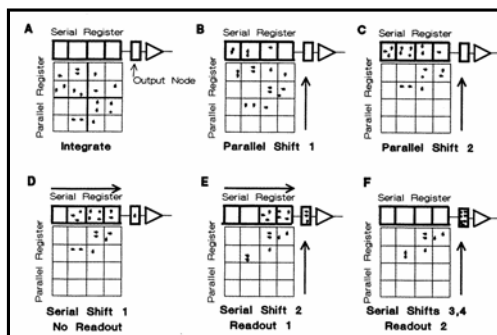
Slow scan camera

- Large area sensor (CCD), square pixels
- Full frame transfer (mechanical shutter)
- non standard video, not real time
- Allows longer integration on sensor
- Higher dynamic range (> 8 bit: 10,12,16 bit)
- ADC integrated in the camera unit
 - photometrics, kodak megapixels, xillix ...
- Special DSP board for data transfer
- binning on chip/sensor

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62

Binning (2x2)

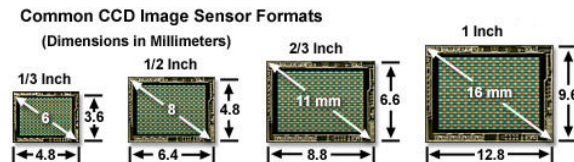


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63

Size of the CCD

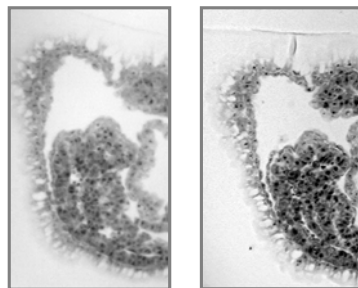
- Size of the CCD sensor contributes to resolution;
- Increase over past 10 years
- More and smaller CCD-wells (pixels)
- Match CCD size to NA, Lens and Measurement



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64

Improvement in CCD digitization



1990

1999

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65

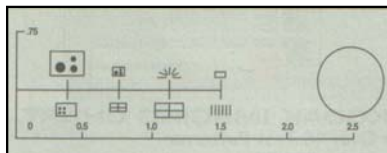
Patterns for Camera Calibration

- Standardized patterns and/or recticles.
- Digitize the pattern and measure.
- This will help you to find the true size of a pixel in an acquisition system.
- A measurement has a deviation!,
model: normal distribution.

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66

Test Slides (1)

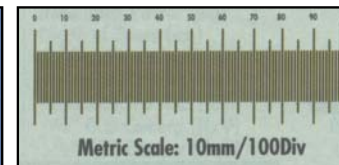


- Circles 2000-250 μm diameter
- Circles 500-62.5 μm diameter
- Scales, Squares etc.

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Test Slides (2)



bars visible at magnification

pixels between two bars

Q: How can we determine the pixel size in x-y using this micrometer.

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68

Review



- Sampling theory
- Applied to microscopy and computed the sampling density for given lens
- Reconstruction
- Aliasing
- CCD camera's
- Mode of operation
- Measurement

Acknowledgements

- FJV experimental results
- Microscopy primer (zeiss, nikon, olympus & leica)
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- Edmund scientific catalogue